Full Stack Assignment – 25

Assignment ID: FSD-25

Title: Data Access Control Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Data Access Control Management System** that allows administrators to assign datasets to users and control access permissions.

The system must ensure users can access **only authorized datasets** and must persist all data in a database.

2️⃣ User Roles

USER

- Register and log in to the system
- View datasets assigned to them
- Access dataset details (read-only)

ADMIN

- Register and log in to the system
- Create and manage datasets
- Assign or revoke dataset access for users
- View all access mappings

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Dataset Management

- ADMIN can create datasets with basic metadata
- Dataset information must be stored in the database
- Dataset visibility must be controlled via access mappings

Access Assignment

- ADMIN assigns datasets to users
- Access assignments must be stored in the database

- USER can view and access only datasets explicitly assigned to them

4️⃣ Business Rule (Critical)

⚠️ **Users must be able to access only datasets assigned to them.**

Any attempt to access unassigned datasets must be rejected at the backend.

5️⃣ Technical Constraints (Mandatory)

- All dataset and access-mapping data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Dataset listing view for USER (assigned datasets only)
- Dataset creation and access assignment view for ADMIN

UI design is not evaluated; authorization correctness is the priority.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **access control correctness, role enforcement, and end-to-end execution**.